

Rösler, J., Bäker, M. & Harders, H., eds (2006), *Mechanisches Verhalten der Werkstoffe*,
SpringerLink Bücher, 2., durchgesehene und erweiterte auflage edn, Teubner, Wiesbaden.

Introduction to Fatigue

Son Tung

Materials Mechanics and Fracture Mechanics, May 2025

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What is Fatigue

Definition

Exposure of material parts under time-dependent **cyclic loads**.

Load **amplitude** is below the material strength.

Initiate at nano scale, micro cracks then **propagate until failure** occurs at macro scale.

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— Fatigue definition

What is Fatigue

What is Fatigue

Definition
Exposure of material to cyclic loading
Load **amplitude** is
Initiate at nano scale

Definition

Exposure of material parts under time-dependent cyclic loads

Load amplitude is below the material strength.

Initiate at nano scale, micro cracks then **propagate** until failure occurs at macro scale

1. An example of fatigue failure was the airline crashes of the De Havilland Comets in 1954. Three of these passenger jets broke up mid-air and crashed within a single year. Sharp corners around the plane's window openings were found to have acted as stress concentrators which initiated cracks. Pressurisation of the aircraft during each flight created stress cycles in the fuselage that propagated the cracks over time. At some point a critical crack length was reached and fast fracture of the fuselage shell then occurred.

<https://www.unsw.edu.au/science/our-schools/materials/engage-with-us/high-school-students-and-teachers/online-tutorials/crack-theory/fatigue-failure/example-fatigue-fracture> 02/05/25

What is Fatigue

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Initiate at nano scale, micro cracks then **propagate until failure** occurs at macro scale.

Responsible for 3 airplane failure in 1954

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Responsible for 3 airplane failure in 1954

Fatigue stages in Metal

Generally, there are 3 stages of crack evolution

- ▶ Initiate (I)
- ▶ Propagate (II)
- ▶ Fracture (III)

Initiate of Fatigue

Assume one has perfect bulk metallic material, meaning no vacancy and completely flat surface. Dislocations are thermally activated and are the crack initiation site.

Stress intensity factor & Intrusion-Extrusion mechanism

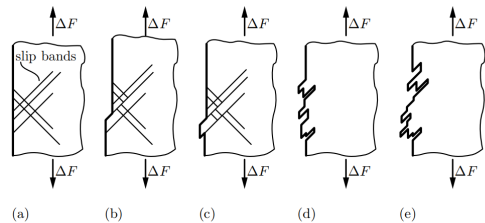


Fig. 10.8. Formation of intrusions and extrusions due to cyclic loading. During the cycles, small irreversible plastic deformations occur on parallel slip bands that cause accumulation of extrusions and intrusions on the surface. The number of cycles increases from the left to the right

$$K = \beta \sigma \sqrt{\pi a} \tag{1}$$

- ▶ Dislocations move on slip planes
- ▶ Extrusions and Intrusions do not return to original position
- ▶ Recess nucleates into microcracks

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Initiate of Fatigue

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1. Beta encapsulate geometry factors, the sharper the crack tips the easier for crack to initiate and propagate
2. The greatest part of the components life time is spent during this initial stage.
3. If the surface is hardened - work hardening or nitriding of steel, dislocation movement near the surface is prevented further. This increases fatigue life.

Propagation of Fatigue I

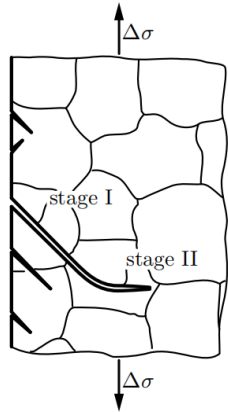


Figure: Transition state of stage I and II Rösler et al. (2006)

- ▶ Cracks initiate at roughly 45° now propagate perpendicular to the stress field (mode I), so called **first propagation phase**
- ▶ Crack growth at the stage transition is relatively slow and might be stopped by obstacles

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Previously opening mode is in mixture state, now mode I dominate

Propagation of Fatigue I



Figure: Transition state of stage I and II Rösler et al. (2006)

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Propagation of Fatigue II

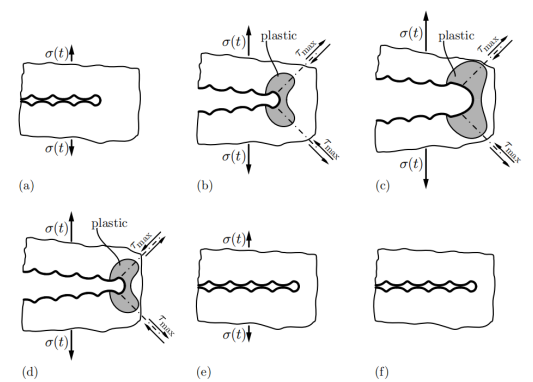
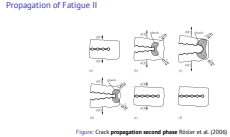


Figure: Crack **propagation second phase** Rösler et al. (2006)

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Panel (a): Minimum Load At the minimum load of the cycle, the crack faces are shown to be in contact, or "closed". This crack closure is a fundamental phenomenon in fatigue (explain later).

Panel (b): Increasing Load and Plastic Zone Formation As the tensile stress increases from equilibrium, crack tip neighbor region experiences stress much higher than the nominal stress applied to the bulk material. When this high local stress exceeds the yield strength of the material, a small region near the crack tip deforms plastically. This shaded area represents the plastic zone. Yielding in ductile materials is driven by shear stresses. Plastic deformation occurs primarily by shear slipping along these directions.

Panel (c): Maximum Load The crack is now fully open, and the plastic zone at the tip is fully developed for this load level. The material within this zone has undergone significant plastic (permanent) deformation.

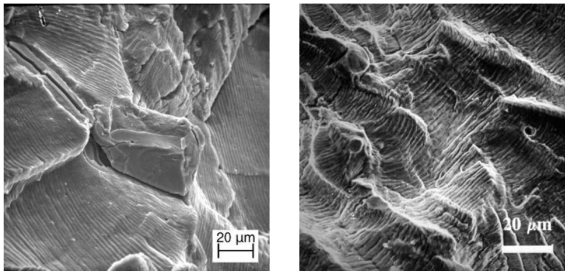
Panel (d): Unloading The elastic material outside the plastic zone tries to recover, but the material within the plastic zone has permanently deformed. This mismatch between the elastic recovery of the bulk material and the permanent deformation in the plastic zone leads to residual stresses and causes the crack faces to push against each other.

Panel (e): Minimum Load and Crack Closure As the load is further reduced to its minimum value, the crack faces come into contact again. Due to the plastic deformation that occurred during the loading phase, the crack faces may even be in contact before the applied stress reaches its absolute minimum, a phenomenon called plasticity-induced crack closure. The residual plastic wake behind the crack tip essentially props the crack open, leading to contact upon unloading.

Panel (f): Crack at Minimum Load (Cycle Complete) The crack remains closed at the minimum load. The cycle is complete, and the crack tip has undergone a sequence of blunting (due to yielding) and potentially re-sharpening or crack advance mechanisms driven by the cyclic plasticity and crack closure effects.

Propagation of Fatigue III

Each loading cycle creates a **striation** within the material, the defect is unique to fatigue failure.



(a) Striations in AlCu 5. The striations are deflected at the Al₂Cu particle [90] (b) Striations in copper

Fig. 10.12. Scanning electron microscope micrograph of striations on two fatigue fracture surfaces

Figure: Striations and Beach marks Rösler et al. (2006)

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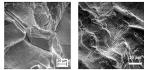
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The grain size also have influence on propagation. The crack propagation rate is smaller in coarse-grained materials than in fine-grained ones.

Also, there are fatigues in other material types: ceramic - too brittle, fail before reaching fatigue, polymer need to take into account heat, composite very complex

Propagation of Fatigue III

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Types of Fatigue materials

Fatigue limit is the lowest stress value below that there exists no fatigue failure.

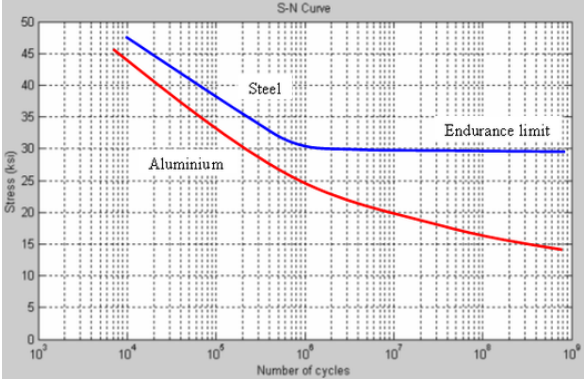
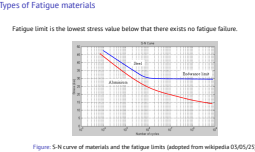


Figure: S-N curve of materials and the fatigue limits (adopted from wikipedia 03/05/25)

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Fatigue limit in metal is directly linked with dislocations irreversible damage. Interstitial atoms in steel are special as they support Cottrell's dislocation, hinder dislocation movement; furthermore, propagation of crack tip. BCC/FCC fatigue limit/no fatigue limit

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Load characteristic

Assume load is periodically cyclic

- ▶ $\sigma_a = (\sigma_{max} - \sigma_{min})/2$
- ▶ $\sigma_m = (\sigma_{max} + \sigma_{min})/2$
- ▶ $R = \sigma_{min}/\sigma_{max}$

R ratio indicates stress type, example
tensile-compress

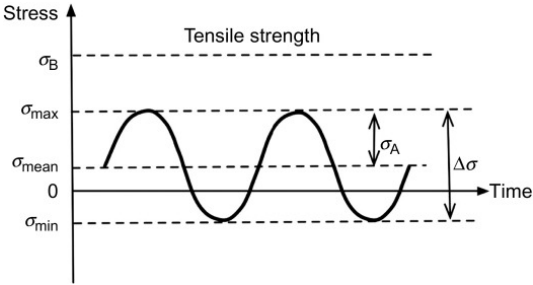


Figure: Typical stress parameters

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Load characteristic

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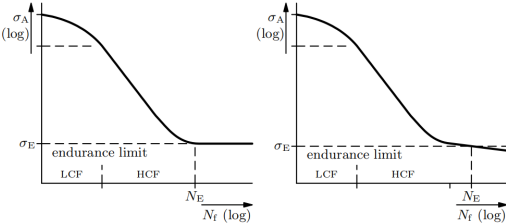
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Table 10.1. Important <i>R</i> ratios				
pulsating in compression	$\sigma > 0$ $\sigma < 0$		$\sigma_{max} < 0$	$R > 1$
zero-to-compression	$\sigma > 0$ $\sigma < 0$		$\sigma_{max} = 0$	$R = -\infty$
reversed	$\sigma > 0$ $\sigma < 0$		$\sigma_m < 0$	$-\infty < R < -1$
fully reversed	$\sigma > 0$ $\sigma < 0$		$\sigma_m = 0$	$R = -1$
reversed	$\sigma > 0$ $\sigma < 0$		$\sigma_m > 0$	$-1 < R < 0$
zero-to-tension	$\sigma > 0$ $\sigma < 0$		$\sigma_{min} = 0$	$R = 0$
pulsating in tension	$\sigma > 0$ $\sigma < 0$		$\sigma_{min} > 0$	$0 < R < 1$
static	$\sigma > 0$ $\sigma < 0$		$\sigma_{min} = \sigma_{max}$	$R = 1$

S-N curve

Known as stress-cycle (S-N) diagram easy to characterize high-cycle fatigue. Strain-cycle (ϵ - N) diagram is less common, essential for low-cycle fatigue testing.

The boundary between High-cycle fatigue (HCF) and Low-cycle fatigue (LCF) is not a clear cut (10^3 to 10^4)



(a) Type I (b) Type II

Fig. 10.19. The characteristic types of S-N curves

Figure: SN diagram Rösler et al. (2006)

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 - S-N curve

Uniaxial s-n testing (video) can be combined with shear from torsion, combined tension/compression from bending, or complex multiaxial states fatigue testing

Both graphs plot cyclic stress amplitude are log scale.

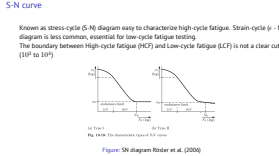
The number of cycles to failure N_f . Stress amplitude at failure σ_A

Higher stress or strain amplitude/range per cycle leads to a shorter fatigue life.

HCF regime, stress amplitude is the dominant factor, LCF regime, plastic strain amplitude is the dominant factor.

S-N curve could be bad: In real-world applications, loading is variable. The sequence, amplitude, and mean stress of each cycle matter. No history is record, meaning hysteresis is not taken into account. Dependency on Geometry and Surface are also ignored.

In alternative, Fatigue Crack Growth (FCG) Rate Curves (e.g., Paris Law), Constant Life Diagrams Haigh Diagrams, Probabilistic S-N Curves. Single temperature. This testing is not complete.



Predict metallic material's life time

Basin equation (valid for HCF)

$$\sigma_A = \sigma_f'(2N_f)^{-a} = R_m N^{-(1/m)}$$

σ_f' : Fatigue strength coefficient

R_m : ultimate tensile stress

Suggest exist relationship between fatigue strength coefficient and material's ultimate tensile stress.

Can roughly estimate S-N HCF regime only with 1 material parameter.

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 - Predict metallic material's life time

Exist strain-life curve but skip due to time constraint

Predict metallic material's life time

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Constant life diagram

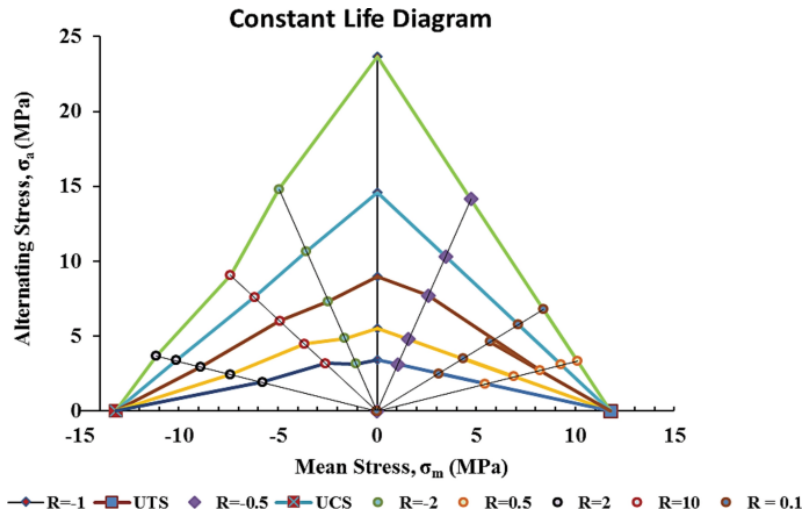


Figure: Plot Stress Amplitude vs. Mean Stress for a fixed life cycle

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- Constant life diagram

<https://link.springer.com/article/10.1007/s11668-022-01585-5> 05/05/25

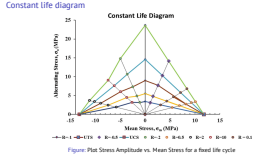


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- ▶ Fatigue is complicatedly describe by various load's characteristic parameters.
- ▶ Understanding crack tip mechanics (plasticity, closure) explains why cracks grow incrementally.
- ▶ Diagrams to describe fatigue is not complete and only partially reflect real situation.
- ▶ Fatigue analysis involves significant complexities and uncertainties due to dependency on material behavior, loading, and geometry.